Predicting Diagnostic Outcomes Using Machine Learning

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1 Problem Formulation

The objective of this study is to predict patient diagnostic outcomes using demographic and clinical variables. The dataset contains measurements related to sleep quality, fatigue severity, stress levels, and behavioural indicators. The target variable *diagnosis* contains three categories: *Depression*, *CI*, and *Both*. Consequently, the task is formulated as a supervised multiclass classification problem.

The dataset initially contained 670 observations and 19 variables. The aim is to determine whether the available demographic and clinical variables can accurately predict diagnostic outcomes.

2 Data Analysis and Feature Selection

Initial exploration confirmed that the dataset contains no missing values and no duplicate observations. The dataset size is relatively small to moderate, making classical machine learning models more appropriate than deep learning approaches.

Variables were classified according to their statistical meaning rather than their storage data types. Nominal variables represent unordered categories, ordinal variables possess a natural ordering, binary variables represent two-category indicators, and numerical variables were further distinguished as discrete or continuous based on their observed distributions and measurement scales. This classification guided subsequent preprocessing decisions such as encoding strategies and feature scaling.

Table 1 summarises the final variable classification used for preprocessing.

Variable Type	Variables
Nominal	diagnosis
Ordinal	social_activity_level, exercise_frequency, work_status
Binary	gender, meditation_or_mindfulness
Discrete Numerical	depression_phq9_score, pem_duration_hours
Continuous Numerical	age, stress_level, sleep_quality_index, fatigue_severity_scale_score
Identifier	id (removed)
Constant Feature	country (removed)

Table 1: Final variable classification used for preprocessing

Inspection of the dataset revealed that the *id* variable represents a unique patient identifier and therefore provides no predictive value. Additionally, the *country* variable was constant across all observations and was removed from the dataset.

A statistical summary of the numerical variables revealed that *depression_phq9_score* contained a value of 99, which lies outside the valid clinical range (0–27) of the PHQ-9 depression scale. This observation was treated as an invalid entry and removed, resulting in a dataset containing 669 observations.

Several visualisation techniques were used to analyse the dataset. Histograms and violin plots were generated to examine distributional properties such as shape, symmetry, and density concentration. Most numerical variables exhibited smooth bounded distributions across their respective scales without evidence of extreme multimodality. In particular, most continuous variables, such as age, sleep quality indices, stress level, brain fog level, and physical pain score, displayed relatively smooth distributions, while discrete variables such as *depression_phq9_score* and *pem_duration_hours* showed step-like patterns consistent with integer-valued measurements.

Boxplots were used to identify potential outliers using the interquartile range criterion. Although some values were detected outside the whisker boundaries for *fatigue_severity_scale_score* and *depression_phq9_score*, these observations remained within the clinically valid ranges specified by the dataset description. Strip plots were further used to confirm the observed minima and maxima of these variables, indicating that these were legitimate observations rather than anomalous measurement errors.

Scatterplots were used to investigate relationships between selected numerical variables. For example, the relationship between stress level and fatigue severity suggested that fatigue severity tends to be higher for the *CI* and *Both* diagnostic groups, while stress levels were distributed more broadly across all classes. This indicates that stress alone is unlikely to be a strong discriminator of diagnosis.

Class distribution analysis revealed moderate class imbalance, with the *Depression* and *CI* classes containing a similar number of observations, whereas the *Both* class contained fewer samples. Although the imbalance is not severe, it is sufficient to influence model evaluation. Figure 1 illustrates the class distribution and motivates the use of stratified sampling in the experimental protocol.

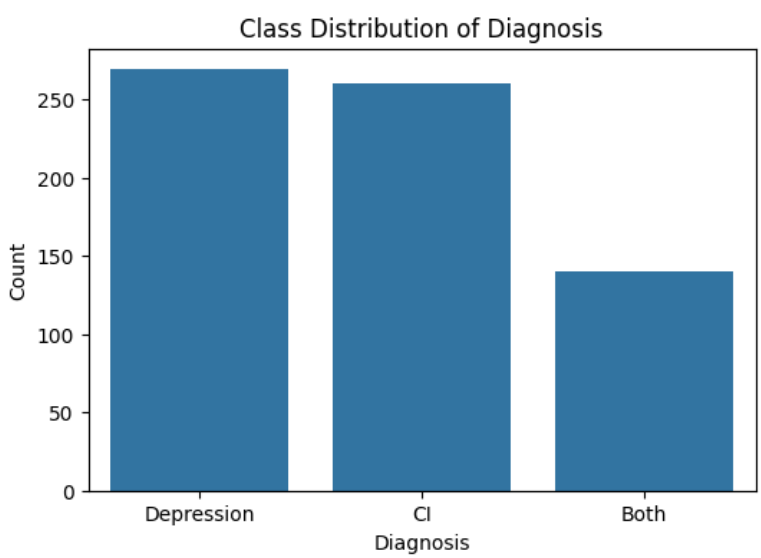

Figure 1: Distribution of diagnosis classes

These exploratory analyses provided insight into the structure of the dataset and informed

subsequent feature selection and modelling decisions.

2.1 Correlation and Redundancy Analysis

A Pearson correlation heatmap was generated to examine relationships among numerical variables, shown in Figure 2. A very strong correlation ($r = 0.97$) was observed between *sleep_quality_index* and *deep_sleep_quality_index*. Since highly correlated variables introduce redundancy and may negatively affect linear models, further analysis was performed.

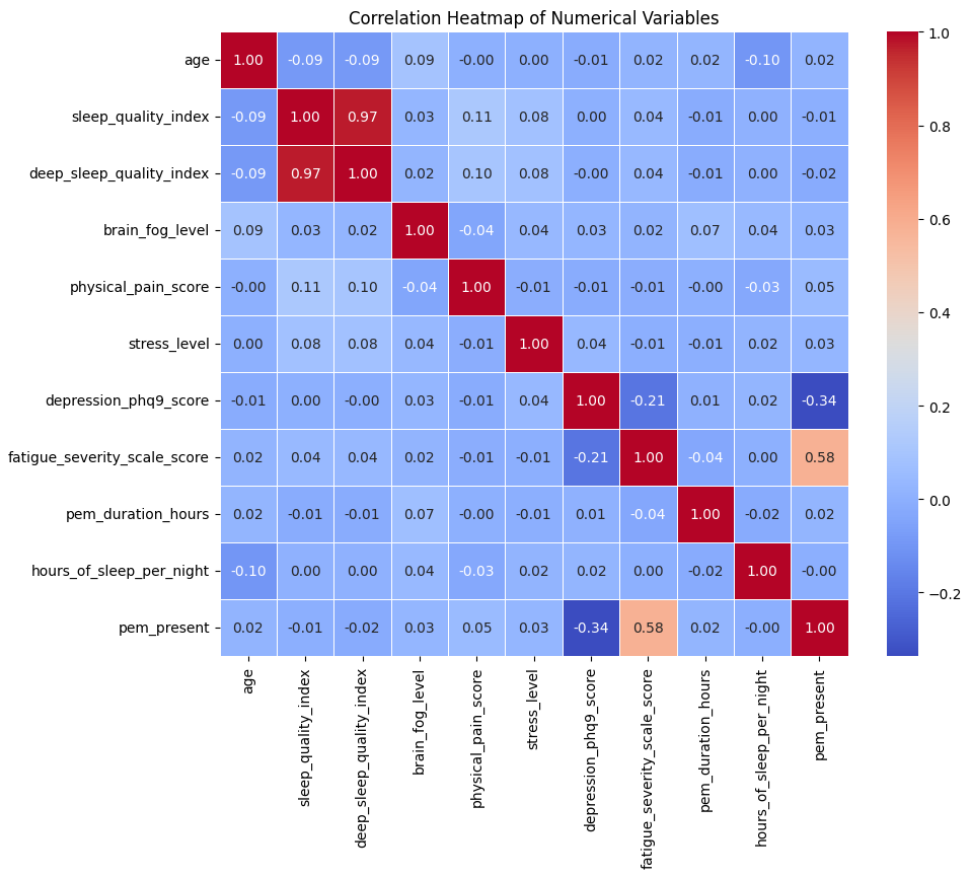

Figure 2: Pearson correlation heatmap of numerical variables before feature reduction

To formally assess whether both sleep-related variables contributed useful discriminatory information, point plots and one-way ANOVA tests were conducted across diagnosis groups. Neither variable showed statistically significant mean differences across the three diagnostic categories. However, *sleep_quality_index* showed marginally stronger group separation and a slightly stronger statistical signal. Therefore, *deep_sleep_quality_index* was removed to reduce multicollinearity while retaining the more informative variable. Aside from this pair, the numerical variables exhibited only weak to moderate correlations, indicating that the remaining features captured largely distinct aspects of patient health.

2.2 Categorical Association Analysis

To examine the relationship between categorical predictors and the target variable (*diagnosis*), Cramér's V statistics were computed. Cramér's V is a measure of association between categorical variables derived from the chi-square statistic and ranges from 0 (no association) to 1 (perfect association). It is commonly used to evaluate the strength of relationships between categorical variables in classification problems.

The computed Cramér's V values between each categorical predictor and the diagnosis variable are presented in Table 2.

Variable	Cramér's V
gender	0.013
work_status	0.082
social_activity_level	0.103
exercise_frequency	0.084
meditation_or_mindfulness	0.029
pem_present	1.000

Table 2: Cramér's V association between categorical predictors and the diagnosis variable

The results indicate negligible to weak associations for most categorical predictors. Specifically, *gender* (0.013) and *meditation_or_mindfulness* (0.029) show almost no association with the diagnosis variable, while *work_status* (0.082), *exercise_frequency* (0.084), and *social_activity_level* (0.103) exhibit weak relationships.

However, the variable *pem_present* exhibits a perfect association with the target variable ($V = 1.000$). Such a value indicates that the feature deterministically partitions the diagnosis categories. Examination of the contingency table confirmed that each value of *pem_present* corresponds exclusively to specific diagnosis classes, with no overlap between categories.

This indicates that *pem_present* encodes target-related information and therefore constitutes data leakage. Including such a variable during model training would artificially inflate predictive performance without reflecting genuine predictive relationships. Consequently, *pem_present* was removed from the feature set prior to model training.

After removing *pem_present*, Cramér's V was computed among the remaining categorical predictors to evaluate pairwise associations between variables. Figure 3 shows that all off-diagonal associations are negligible ($V < 0.10$), indicating no evidence of redundancy or deterministic relationships among the retained categorical variables.

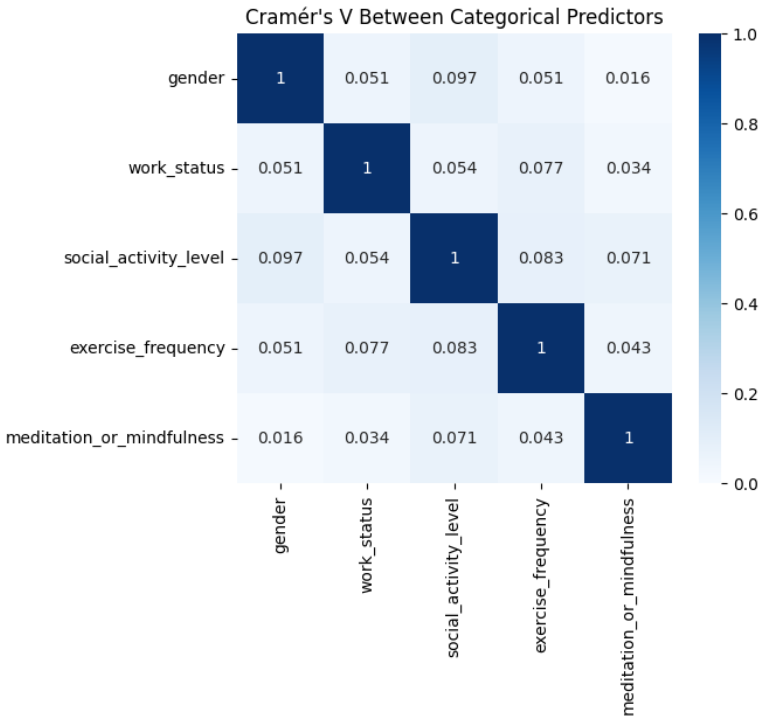

Figure 3: Cramér's V heatmap between categorical predictors after removing *pem_present*

Overall, the feature screening process removed the identifier column, the constant feature, the invalid PHQ-9 observation, the highly redundant sleep-related variable, and the leakage-prone *pem_present* variable. The resulting feature set was considered suitable for modelling.

3 Experimental Protocol

To ensure reliable model evaluation, the dataset was divided into training and testing subsets using an 80/20 stratified train–test split. Stratification preserves class distribution across both subsets.

Model selection and hyperparameter tuning were performed using stratified 5-fold cross-validation on the training set. Macro F1-score was selected as the primary evaluation metric because it provides balanced performance across all classes and is robust to moderate class imbalance.

Machine learning models require numerical inputs; therefore, categorical variables were encoded prior to model training. Nominal variables such as *gender* and *work_status* were encoded using one-hot encoding to avoid imposing artificial ordinal structure. Ordinal variables such as *social_activity_level* and *exercise_frequency* were encoded using ordinal encoding that preserved their inherent ordering. Binary variables were mapped to 0/1 representation. Numerical variables were standardised for models that are sensitive to feature magnitude, such as Logistic Regression and k-Nearest Neighbours, while tree-based models were trained without scaling.

To prevent data leakage, encoding and scaling were implemented within a preprocessing pipeline and applied during cross-validation rather than to the full dataset prior to splitting.

4 Modelling Framework

This section summarises the modelling approach, preprocessing decisions, experimental protocol, selected machine learning methods, and hyperparameter tuning strategy used in this study.

Problem Modelling. The task was formulated as a supervised multiclass classification problem. The objective is to predict the patient diagnosis category (*Depression*, *CI*, or *Both*) using demographic, behavioural, and clinical features. Since the target variable contains three discrete classes, classification algorithms capable of handling multiclass outputs were required.

Decisions Based on Data Analysis. Exploratory data analysis guided several preprocessing decisions. The identifier column *id* was removed because it does not contain predictive information. The variable *country* was removed as it was constant across all observations. An invalid value in *depression_phq9_score* (99) was identified and removed based on the known clinical range of the PHQ-9 scale. Correlation analysis revealed a strong relationship between *sleep_quality_index* and *deep_sleep_quality_index*, and the latter was removed to reduce redundancy. Cramér's V analysis identified *pem_present* as a leakage variable with perfect association with the target variable, and it was therefore removed to prevent artificially inflated model performance.

Experimental Protocol. To ensure reliable model evaluation, the dataset was divided into training and testing subsets using an 80/20 stratified train–test split. Stratification preserves the class distribution of the target variable across both subsets. Model selection and hyperparameter tuning were performed using stratified 5-fold cross-validation on the training set. Macro F1-score was used as the primary evaluation metric to ensure balanced performance across classes.

Selected Machine Learning Methods. Three machine learning algorithms discussed during the lectures were evaluated: Logistic Regression, k-Nearest Neighbours (KNN), and Random Forest. Logistic Regression provides a linear baseline classifier, KNN represents a distance-based learning approach relying on local neighbourhood similarity, and Random Forest is an ensemble tree-based method capable of capturing nonlinear relationships and feature interactions. Comparing these models allows investigation of whether linear decision boundaries, local instance similarity, or nonlinear ensemble structures best explain the diagnostic outcomes.

Hyperparameter Search Protocol. Hyperparameter tuning was performed using grid search combined with stratified 5-fold cross-validation. For Logistic Regression, the regularisation strength parameter C was tuned. For KNN, the number of neighbours k was varied to control the degree of smoothing in local classification decisions. For Random Forest, the number of trees and maximum tree depth were tuned to balance model complexity and generalisation performance. The configuration achieving the highest cross-validated Macro F1-score was selected for each model.

5 Machine Learning Methods

Three classification algorithms discussed during lectures were selected for comparison:

- Logistic Regression
- k-Nearest Neighbours (KNN)
- Random Forest

These models were chosen because they represent distinct machine learning paradigms.

Logistic Regression is a linear classifier that models the relationship between predictors and class probabilities. It provides an interpretable baseline and is useful for assessing whether the problem can be explained adequately by approximately linear decision boundaries.

KNN is a distance-based instance learning method that classifies an observation using the labels of nearby training examples. It was selected to investigate whether local neighbourhood similarity in the feature space is informative for diagnosis prediction.

Random Forest is an ensemble tree-based method that combines multiple decision trees to capture nonlinear relationships and higher-order feature interactions. It is well suited for structured tabular datasets and often performs strongly when complex dependencies are present.

Comparing these three models therefore allows evaluation of whether linear structure, local similarity, or nonlinear ensemble interactions best explain the diagnostic outcomes.

6 Hyperparameter Search

Hyperparameter tuning was performed using grid search within stratified 5-fold cross-validation. Stratification ensures that each fold preserves the class distribution of the target variable, which is important given the moderate class imbalance observed in the dataset. This approach provides a reliable estimate of model performance while preventing overfitting to a single validation split. Table 3 summarises the search spaces and best configurations obtained during model selection.

Model	Hyperparameter Grid	Best Hyperparameters	CV Macro F1
Logistic Regression	$C \in \{0.1, 1, 10\}$	$C = 10$	0.774
k-Nearest Neighbours	$k \in \{3, 5, 7, 9, 11\}$	$k = 11$	0.548
Random Forest	$n_estimators \in \{200, 500\}$, $max_depth \in \{None, 10, 20\}$	$n_estimators = 500$, $max_depth = 10$	0.947

Table 3: Hyperparameter tuning results using grid search with stratified 5-fold cross-validation

The hyperparameter ranges were selected to balance model flexibility and computational efficiency. For Logistic Regression, the search focused on the regularisation strength parameter C. For KNN, the number of neighbours was tuned in order to control the degree of smoothing in local decision-making. For Random Forest, the grid explored both the ensemble size and the maximum tree depth.

The hyperparameter search results presented in Table 3 show clear differences in model performance. For Logistic Regression, the optimal value of the regularisation strength parameter was $C = 10$, producing a cross-validated Macro F1-score of 0.774. This suggests that a relatively lower level of regularisation allowed the linear model to better capture relationships between predictors and diagnostic outcomes.

For the k-Nearest Neighbours model, the best-performing configuration used $k = 11$, yielding a Macro F1-score of 0.548. The comparatively lower performance indicates that local neighbourhood similarity alone was insufficient to adequately separate the diagnostic classes.

The Random Forest model achieved the highest cross-validated Macro F1-score of 0.947, with an optimal configuration of $n_estimators = 500$ and $max_depth = 10$. This strong result suggests that nonlinear relationships and feature interactions play a substantial role in predicting diagnosis. Based on these results, Random Forest was selected as the final candidate model for evaluation on the unseen test set.

7 Model Comparison

Table 4 summarises the cross-validated Macro F1-scores of the three candidate models.

Model	Cross-Validated Macro F1 Score
Logistic Regression	0.774
k-Nearest Neighbours	0.548
Random Forest	0.947

Table 4: Cross-validated model comparison

Random Forest substantially outperformed both Logistic Regression and KNN. This suggests that nonlinear relationships and feature interactions are important for predicting diagnosis. Logistic Regression achieved moderate performance, indicating that some linear structure is present in the data. In contrast, the lower performance of KNN suggests that local neighbourhood similarity alone is insufficient to capture the more complex patterns contained within the dataset.

Based on these results, Random Forest was selected as the final candidate model for evaluation on the unseen test set.

8 Final Model Evaluation

The selected Random Forest model was evaluated on the held-out 20% test set. The final model achieved strong predictive performance, with a test accuracy of approximately 0.963 and a Macro F1-score of approximately 0.954.

Table 5 summarises the final test metrics.

Metric	Value
Test Accuracy	0.963
Test Macro F1	0.954

Table 5: Final performance of the selected Random Forest model on the test set

Figure 4 provides the full classification report. The *CI* category achieved perfect precision, recall, and F1-score. The *Depression* category also achieved strong performance. The *Both* category exhibited slightly lower recall, indicating that a small number of instances were misclassified.

Classification Report

	precision	recall	f1-score	support
Both	0.96	0.857	0.906	28.0
CI	1.0	1.0	1.0	52.0
Depression	0.93	0.981	0.955	54.0
accuracy	0.963	0.963	0.963	0.963
macro avg	0.963	0.946	0.954	134.0
weighted avg	0.963	0.963	0.962	134.0

Figure 4: Classification report for the final Random Forest model

Figure 5 shows the confusion matrix for the final Random Forest classifier. The matrix indicates that the *CI* class was perfectly classified, while one *Depression* observation and four *Both* observations were misclassified. Most errors occur between the *Both* and *Depression* classes, suggesting some overlap in their clinical profiles.

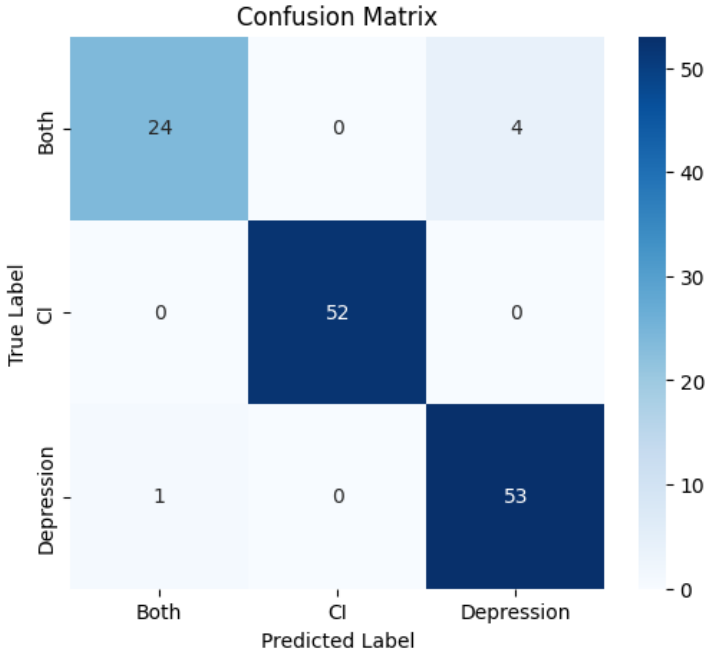

Figure 5: Confusion matrix of the final Random Forest classifier

The consistency between cross-validation performance and held-out test performance suggests that the model generalises well and does not exhibit substantial overfitting.

9 Conclusion

This study investigated the use of machine learning techniques to predict diagnostic outcomes based on patient demographic and clinical variables. Through comprehensive exploratory data analysis, feature screening, and systematic model comparison, Random Forest was identified as the most effective algorithm.

The final model achieved strong predictive performance on unseen data, demonstrating that ensemble tree-based methods are well suited for structured healthcare datasets in which nonlinear relationships and feature interactions are present.

Future work could explore additional ensemble methods, alternative feature engineering strategies, or larger datasets in order to further improve predictive performance.